gpeyt1-09 · Falling-head k

GEO PE · Long Calculation Problems I — Topic 2: Soil Mechanics, Lab Testing & Analysis

Problem

A falling-head permeability test is run on a silty sand specimen 2.5 in. in diameter and 6.0 in. long. The standpipe inside diameter is 0.25 in. The head drops from 40 in. to 24 in. in 26 min. Water temperature 20°C; specimen dry unit weight 104 pcf; Gs = 2.66.

Most nearly, the hydraulic conductivity k is:

A. 5.0e-05 cm/s
B. 2.2e-05 cm/s
C. 2.0e-05 cm/s
D. 5.0e-04 cm/s

Given equations / data

> k = (a L)/(A t) · ln(h1/h2); a = standpipe area, A = specimen area

Worked solution

1. a = π/4 (0.25)² = 0.04909 in.²; A = π/4 (2.5)² = 4.9087 in.²
2. t = 26 × 60 = 1560 s
3. k = (0.04909×6.0)/(4.9087×1560) × ln(40/24) = 1.965e-05 in./s
4. × 2.54 = 4.99e-05 cm/s

Final answer

5.0e-05 cm/s

Why the other values are wrong

- **2.2e-05 cm/s** — used log10 instead of ln
- **2.0e-05 cm/s** — left the result in in./s but labeled cm/s
- **5.0e-04 cm/s** — used diameter ratio d/D instead of area ratio a/A

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.8.5 Permeability Testing, p. 139

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=RqaahrVtV4k (no creator values copied).

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